MALIS Project 1: Multi-Armed Bandits

Tasks 2 & 3 Report: Human Play Analysis and Performance Isolation

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1 Task 2: Human Play Analysis and Strategy Formalization

1.1 Gameplay Comparison with Strategy Notes

Only one team member (Thomas Decker, `fishing_log_decker.json`) completed the Moodle activity before the deadline. Across all three seasons, his actions strictly matched his written strategy notes (Table 1):

Season	Days	Explored	Committed Spot	Raw Catch	Oracle	Emp. Ratio (ρ)	Exp. Ratio (η)
Season 1	30	Days 1–5	Spot 0 (Old Pier)	260.6 kg	300.0 kg	86.87%	92.67%
Season 2	10	Days 1–5	Spot 0 (Old Pier)	71.5 kg	99.0 kg	72.22%	80.10%
Season 3	30	Days 1–5	Spot 3 (Deep Channel)	270.4 kg	271.5 kg	99.60%	92.78%

Table 1: Summary of human gameplay vs. oracle targets and noise-free efficiency (η).

- **Season 1 (30 days):** The player tested all 5 spots once during days 1–5 (spots $0, 2, 4, 3, 1$, yielding $9.0, 5.5, 1.8, 2.0, 0.0$ kg) and exploited spot 0 exclusively for days 6–30, matching his note: *"I tried all the fish spot and one had like the double amount [...] so I chose to always go there."*
- **Season 2 (10 days):** He repeated the exact 5-day round-robin exploration (spots $0, 2, 1, 3, 4$, yielding $10.1, 1.0, 3.6, 0.2, 8.4$ kg) and exploited spot 0 for days 6–10 (*"Same strategy than previously"*).
- **Season 3 (30 days, shift at day 16):** After exploring spots $0, 2, 1, 3, 4$ on days 1–5 ($2.2, 5.2, 1.5, 10.4, 3.0$ kg), he committed to spot 3 for days 6–30 without ever re-exploring (*"As always I did the same strategy"*).

1.2 Formal Algorithmic Rule (1-Sample Explore-Then-Commit)

For any bay with $K = 5$ spots over T days ($T \geq K$), let $A_t \in \{0, \ldots, K-1\}$ be the spot chosen on day t and R_t the observed catch:

1. **Exploration ($1 \leq t \leq K$):** Sample each spot once without replacement ($A_t = t - 1$) and record the initial catch $r_{A_t} = R_t$.
2. **Commitment ($K < t \leq T$):** Select the empirically best spot $a^* = \arg\max_a r_a$ (breaking ties randomly) and play $A_t = a^*$ for all remaining days.

2 Task 3: Separating Decisions from Luck & Season Comparison

2.1 Performance Measure Derivation

Raw total catch $R_{\text{total}} = \sum_{t=1}^{T} R_t$ conflates decision quality with two sources of luck in `fishing.py`: (1) **bay richness**, since spot means $\mu_t(a)$ are drawn uniformly in $[2.0, 10.0]$ kg, and (2) **daily Gaussian noise**, since $R_t = \max(0, \mu_t(A_t) + \sigma\epsilon_t)$ with $\sigma = 2.5$ kg.

Using `bay.best_possible()` and `bay.reveal()`, where $\mu_t^* = \max_a \mu_t(a)$, we define two normalized metrics:

$$\rho_{\text{empirical}} = \frac{\sum_{t=1}^{T} R_t}{\sum_{t=1}^{T} \mu_t^*}, \qquad \eta_{\text{expected}} = \frac{\sum_{t=1}^{T} \mu_t(A_t)}{\sum_{t=1}^{T} \mu_t^*} = 1 - \frac{\text{Regret}_{\text{expected}}}{\sum_{t=1}^{T} \mu_t^*} \tag{1}$$

While $\rho_{\text{empirical}}$ removes bay richness disparity, η_{expected} also eliminates daily Gaussian noise σ, measuring purely the expected value of the chosen actions. Running `RandomAgent` across 500 random Season 1 bays confirms that relative dispersion ($CV = \text{SD}/\text{Mean}$) drops from 19.18% (raw catch) to 15.96% ($\rho_{\text{empirical}}$) and 14.55% (η_{expected}), achieving a 24.1% variance reduction (Table 2).

Metric (500 Season 1 Bays, RandomAgent)	Mean	SD	Coeff. of Variation (CV)
1. Raw Total Catch (R_{total})	180.3 kg	34.6 kg	19.18%
2. Empirical Performance Ratio ($\rho_{\text{empirical}}$)	0.696	0.111	15.96%
3. Expected Efficiency Ratio (η_{expected})	0.694	0.101	**14.55%**

Table 2: Variance reduction across 500 random bays under a fixed decision policy.

2.2 Cross-Season Comparison

- **Shorter Horizon in Season 2 ($T = 10$):** Spending 5 days exploring consumed 50% of the season (vs. 16.7% in Season 1). Consequently, expected efficiency η dropped from 92.67% to 80.10% ($\rho = 72.22\%$), showing that static exploration is too costly on short horizons.
- **Environmental Shift in Season 3 ($t_{\text{shift}} = 16$):** True spot means changed at day 16 from $[2.6, 6.1, 5.6, \mathbf{9.5}, 4.1]$ to $[7.5, 5.4, 5.0, \mathbf{8.6}, 3.8]$. The player did not adapt at all, staying on spot 3 for all 25 exploitation days. Yet, he achieved $\eta = 92.78\%$ (and a lucky $\rho = 99.60\%$ due to positive daily noise) purely because spot 3 happened to remain the optimal arm after the shift ($\mu_3 : 9.5 \to 8.6$).